DIgSILENT PowerFactory

A 12-Week Concept-and-Practice Learning Programme

Power-System Theory, Modelling, Load Flow, Short Circuit, Contingency Analysis, RMS Dynamics and Renewable Integration

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Designed for a study commitment of three hours per week for twelve weeks. The programme deliberately couples every theoretical concept with an implementation task in DIgSILENT PowerFactory so that software operation and engineering interpretation are learned together.

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Chapter 1

What PowerFactory Can Do

1.1 What is DIgSILENT PowerFactory?

DIgSILENT *PowerFactory* is an integrated power-system modelling, analysis and simulation environment used for generation, transmission, distribution, industrial networks and renewable-energy studies. Its central idea is that a single consistent network model can be reused across many types of engineering calculations. A busbar, transformer, line, generator, load, protection device or converter is not merely a symbol on a drawing: it is an object containing electrical parameters, operating data, connectivity, states and calculation results.

The software therefore combines four activities that should always be distinguished mentally:

1. **Network modelling**: describing what physical equipment exists and how it is connected.
2. **Operating-point definition**: specifying which equipment is in service, how much load is demanded, how generators are dispatched, tap positions, control targets and other operational data.
3. **Calculation**: applying an analysis method such as load flow, short circuit, contingency analysis or time-domain simulation.
4. **Engineering interpretation**: deciding whether voltages, currents, loadings, fault levels, dynamic trajectories and protection responses are acceptable.

DIgSILENT describes the PowerFactory base package as including load-flow analysis, short-circuit analysis, sensitivities/distribution factors, basic MV/LV analysis, network representation, model management, outage management, results and reporting, while additional licensed modules extend the software into areas such as contingency analysis, protection, quasi-dynamic simulation, stability, EMT, reliability, probabilistic analysis and optimisation [1, 2].

1.2 The "one model, many studies" philosophy

A common beginner mistake is to treat every calculation as a separate software exercise. In professional studies, the valuable asset is the **network model**. Once that model is trustworthy, many questions can be asked of it.

For example, consider a 110/33-kV substation. The same transformer, line and load data can be used to ask:

- What are the bus voltages at peak demand?
- Is either transformer overloaded if one transformer trips?
- What is the three-phase fault current at the 33-kV bus?

- Does a circuit breaker have adequate interrupting capability?
- What happens to voltage after a nearby fault?
- Can a generator remain in synchronism after fault clearance?
- Does a new solar farm create reverse power flow or overvoltage?
- What reactive-power capability is required from the solar farm?
- Do protection devices coordinate correctly?
- How does the network behave over a full year of variable demand and generation?

This is why careful modelling is more important than memorising menu locations.

1.3 Core analysis capabilities

1.3.1 Load-flow analysis

Load flow computes the steady-state operating point of a power system. Typical outputs are:

- voltage magnitude and phase angle at buses;
- active and reactive power through lines and transformers;
- current and thermal loading;
- generator active/reactive outputs;
- system losses;
- transformer tap positions and controller actions;
- voltage and loading limit violations.

Load flow is the foundation of almost every planning study because it answers: *Can the network carry the required power while maintaining acceptable voltage and equipment loading?*

PowerFactory also provides dedicated MV/LV feeder assessment tools such as voltage profiles and feeder-oriented calculations [4].

1.3.2 Short-circuit analysis

Short-circuit analysis calculates currents and related quantities during electrical faults. Typical studies include:

- three-phase faults;
- single-line-to-ground faults;
- line-to-line faults;
- line-to-line-to-ground faults;
- maximum and minimum fault levels;
- contribution from individual sources;
- breaker duty and equipment withstand assessment.

This analysis is essential for switchgear specification, earthing studies, protection design and verifying that equipment can withstand and interrupt faults.

1.3.3 Contingency and security analysis

Contingency analysis asks what happens after one or more components are lost. The most common planning criterion is $N - 1$: the system should continue to operate acceptably after the loss of one credible component. Advanced PowerFactory contingency functionality supports $n - 1$, $n - 2$ and $n - k$ outage levels, screening and detailed assessment [2].

Typical questions include:

- Does another line overload after Line 1 trips?
- Does a bus fall below its voltage limit after a transformer outage?
- Is load disconnected?
- Can generation be redispatched to remove violations?

1.3.4 Sensitivity and distribution-factor analysis

Sensitivity methods quantify how a system variable changes when another variable is changed. Examples include:

- change in line loading caused by a generator redispatch;
- change in bus voltage caused by reactive-power injection;
- power-transfer distribution factors;
- identifying effective locations for voltage support.

These methods are valuable because they show not only *that* a violation exists, but also which control action is most effective.

1.3.5 RMS time-domain simulation and stability

A load flow contains no explicit time axis. RMS simulation studies how the system evolves after disturbances. It can represent:

- faults and their clearance;
- generator trips;
- line switching;
- load steps;
- governor and AVR response;
- converter controls;
- frequency response;
- rotor-angle stability;
- voltage recovery;
- oscillations.

This is where differential equations, control systems and electromagnetic-mechanical dynamics become important.

1.3.6 Electromagnetic-transient (EMT) simulation

EMT simulation resolves faster electromagnetic phenomena than conventional positive-sequence RMS studies. It is useful for detailed switching transients, power-electronic controls, converter interactions, waveform phenomena and very fast dynamics. PowerFactory 2026 includes further EMT performance developments, including parallelisation based on network partitioning [3].

RMS and EMT are not competitors; they answer different questions. RMS is usually preferred for large-system electromechanical stability when waveform-level detail is unnecessary. EMT is selected when high-frequency and switching detail matters.

1.3.7 Protection analysis

With the relevant licences, PowerFactory can model protection relays, instrument transformers, fuses and circuit breakers; assess protection settings; produce coordination diagrams; and trace fault-clearing sequences. The 2026 advanced-function specification describes support for overcurrent, distance, voltage and frequency protection workflows and short-circuit trace functionality [2].

1.3.8 Harmonic and power-quality analysis

Power-electronic converters, nonlinear industrial loads and resonance can create waveform distortion. Harmonic analysis can be used to study:

- harmonic voltages and currents;
- frequency-dependent impedance;
- resonance;
- filter performance;
- distortion propagation.

This becomes increasingly important in networks with inverter-based resources and large nonlinear loads.

1.3.9 Quasi-dynamic and time-series simulation

A quasi-dynamic simulation repeatedly solves steady-state calculations across time. Instead of studying milliseconds or seconds, the time scale may be minutes, hours, days or years. The advanced-function specification includes time-profile support and medium- to long-term simulation based on steady-state analysis [2].

Typical applications are:

- daily PV generation;
- annual demand profiles;
- transformer tap operations over time;
- battery state of charge;
- renewable curtailment;
- thermal utilisation and energy losses.

1.3.10 Renewable and distributed-generation studies

PowerFactory can represent renewable generation in transmission, distribution and industrial networks. Typical engineering questions include reverse power flow, voltage rise, varying fault level, reactive-power capability, voltage control and equipment loading. DIgSILENT identifies distributed generation and renewable integration as major PowerFactory application areas [6].

1.3.11 Reliability, restoration and probabilistic analysis

With advanced modules, PowerFactory can evaluate reliability indices, component failure models, restoration actions and probabilistic input distributions. The 2026 product specification lists state enumeration, common reliability indices, restoration analysis, Monte Carlo/Quasi-Monte Carlo techniques and probabilistic load-flow-related assessment [7].

1.3.12 Network representation and data management

PowerFactory supports radial and meshed AC/DC networks and detailed substation representations. DIgSILENT states that it supports one-, two-, three- and four-wire arrangements as well as detailed sites, substations, bays and switching equipment [5]. This capability matters because professional network studies often depend as much on correct topology, switching status and data governance as on the numerical algorithm itself.

1.4 What this 12-week programme will and will not cover

The present programme focuses on the highest-value foundation:

Model → **Load Flow** → **Voltage Control** → **Short Circuit** → $N-1$ → **RMS Dynamics** → **Renewable Integration**

Detailed protection coordination, harmonics, EMT, scripting, optimal power flow, reliability, probabilistic studies and advanced small-signal stability are intentionally deferred. They are excellent second-stage topics, but learning them before you can build and diagnose a trustworthy steady-state network is inefficient.

Chapter 2

Training Network Used Throughout the Programme

2.1 Purpose

A single educational network is used for the entire programme. Each week adds complexity or asks a new engineering question. This avoids the common problem of spending most of the available time drawing unrelated tutorial systems.

2.2 Single-line concept

Use the following conceptual topology:

External Grid 220 kV
↓
T1: 220/110 kV
↓
Bus 110-A
↔ Line L1 and Line L2 ↔
Bus 110-B
↓
T2: 110/33 kV
↓
Bus 33-A – Feeder – Bus 33-B
↓ ↓
Loads Future PV Plant

A synchronous generator is connected at 110 kV or 33 kV depending on the particular exercise.

2.3 Suggested educational data

These values are not intended to reproduce a particular utility system. They are selected to give plausible results and to make parameter changes visible.

Object	**Rating / nominal data**	**Suggested starting parameters**
External Grid	220 kV	Short-circuit power $S_k'' = 5000$ MVA; $R/X = 0.1$

T1	250 MVA, 220/110 kV	$u_k = 12\%$, copper component approximately 0.5%, OLTC if desired
L1	110 kV, 40 km	$R' = 0.08\,\Omega/\text{km}$, $X' = 0.35\,\Omega/\text{km}$; select a plausible current rating
L2	110 kV, 55 km	Same type initially; different length to create unequal flows
T2	100 MVA, 110/33 kV	$u_k = 10\%$, OLTC on HV side or according to selected transformer model
Load A	33 kV	25 MW, PF 0.95 lagging
Load B	33 kV	15 MW, PF 0.92 lagging
Load C	110 kV	20 MW, PF 0.97 lagging
Generator G1	50 MVA	Start around 25 MW; voltage-controlled when studying PV buses
PV plant	30 MW	Add in Week 11; compare PF, fixed-Q and voltage-control strategies

Important modelling note

The exact fields available in *PowerFactory* vary with the chosen element model, software version and licence. Treat the numerical dataset as a learning scaffold. If a field is represented differently in your installed version, preserve the physical meaning rather than forcing the exact menu sequence described here.

2.4 How to use the PowerFactory practical guides

The practical instructions in the following chapters are written for the modern PowerFactory desktop interface (PowerFactory 2024–2026 terminology). Minor wording and toolbar placement can differ with software release, workspace layout and licensed modules. When an icon is not visible, use the named command through the main menu, the active Study Case or the Data Manager. The electrical parameter is more important than the colour or position of the icon.

2.4.1 Paths and object names used in this book

A path such as

```
Data Manager -> Network Model -> Network Data -> TrainingGrid
```

means open each item from left to right. Common PowerFactory object classes are useful when searching in the Data Manager:

Object	PowerFactory class
Terminal / bus	`ElmTerm`
External Grid	`ElmXnet`
Two-winding transformer	`ElmTr2`
Line / cable	`ElmLne`
General Load	`ElmLod`
Synchronous Machine	`ElmSym`
Static Generator	`ElmGenstat`
Shunt	`ElmShnt`
Load Flow command	`ComLdf`
Short-Circuit command	`ComShc`
Initial Conditions command	`ComInc`
Run Simulation command	`ComSim`
Short-Circuit Event	`EvtShc`

2.4.2 Study Case versus Operation Scenario

This distinction is used deliberately throughout the labs.

Study Case: stores the calculation context—calculation commands and their options, selected result objects, plots, simulation events and the active operating context. When a calculation is run for the first time, PowerFactory can create the corresponding command object in the active Study Case so the settings can be reused.

Operation Scenario: stores operational settings of the network, such as load set-points, generator dispatch/control values, transformer tap positions and other operational data. Use separate scenarios for light load, peak load, low-load/high-PV and similar conditions instead of repeatedly overwriting a single operating point.

For this course, create scenarios through `File -> Save Operation Scenario as...` when available, or through `Data Manager -> Operation Scenarios -> right-click -> New -> Operation Scenario`. Activate a scenario by right-clicking it and selecting `Activate`; save changed values with `File -> Save Operation Scenario`.

Important modelling note

PowerFactory fields are context sensitive. For example, selecting a different generator control mode changes which P/Q/V/PF fields are editable. If a field named below is not visible, first check the selected control mode and the Basic Data / Load Flow / Short-Circuit / RMS-Simulation tabs. Hovering a field or pressing the field-help button is also useful for confirming the parameter meaning.

2.5 Working convention for the training project

Use the following names exactly. Consistent naming makes the later instructions much easier to follow:

- buses: `Bus220`, `Bus110A`, `Bus110B`, `Bus33A`, `Bus33B`;
- external grid: `GridSource`;
- transformers: `T1`, `T2`;
- lines: `L1`, `L2`, `L33`;

- loads: `LoadA`, `LoadB`, `LoadC`;
- synchronous generator: `G1`;
- solar plant: `PV_30MW`.

Chapter 3

Week 1 – Power-System Structure, Single-Line Diagrams and PowerFactory Orientation

Learning outcomes

By the end of Week 1 you should be able to explain what a bus, line, transformer, load, generator and external grid represent; distinguish a drawing from an electrical model; create a project and grid; place and connect basic objects; and understand the role of a Study Case.

3.1 Three-hour breakdown

0:00–0:50	Theory: three-phase networks, SLDs, electrical power quantities and network topology.
0:50–1:05	Hand exercise: calculate three-phase current from MW, voltage and power factor.
1:05–2:30	PowerFactory: project structure, grid, buses, external grid, transformer, line and load.
2:30–2:50	Experiment: open a connection / switch an element out of service and interpret the effect.
2:50–3:00	Record five concepts learned and three interface actions you can now perform without instructions.

3.2 Theory: how a three-phase power system is organised

An electric power system transfers energy from sources to consumers through networks operating at different voltages. A simplified chain is:

$$\text{Generation} \rightarrow \text{Step-up transformer} \rightarrow \text{Transmission} \rightarrow \text{Substation} \rightarrow \text{Distribution} \rightarrow \text{Load.}$$

Electrical power is transmitted at high voltage because for a given active power,

$$P = \sqrt{3} V_L I_L \cos\phi,$$

where V_L is line-to-line RMS voltage, I_L is line current and $\cos\phi$ is power factor. If P and power factor are fixed, raising V_L reduces current. Resistive loss is

$$P_{\text{loss}} = 3I^2R,$$

so current reduction has a squared benefit.

3.2.1 Active, reactive and apparent power

For sinusoidal steady state, complex power is

$$S = P + jQ,$$

with magnitude

$$|S| = \sqrt{P^2 + Q^2}.$$

P is active power in watts; Q is reactive power in var; and $|S|$ is apparent power in VA. Power factor is

$$\text{PF} = \frac{P}{|S|} = \cos\phi.$$

For a balanced three-phase system,

$$P = \sqrt{3}VI\cos\phi, \qquad Q = \sqrt{3}VI\sin\phi, \qquad |S| = \sqrt{3}VI.$$

Reactive power should not be thought of as “useless power”. It is associated with electric and magnetic field energy and is fundamental to voltage control and AC network operation.

3.2.2 What a single-line diagram means

A single-line diagram compresses a three-phase system into a readable representation. It communicates **topology**: what is electrically connected to what.

Important objects are:

Bus/terminal: a node at which connected equipment shares an electrical voltage.

Line/cable: an impedance path transferring current and power between buses.

Transformer: transfers power between voltage levels and contributes series impedance; taps may regulate voltage.

Load: consumes active and usually reactive power.

Generator: injects active power and may control voltage by adjusting reactive power.

External grid: represents the rest of a large system by imposing/reference-controlling voltage and angle and, for fault studies, an equivalent source impedance.

Switch/circuit breaker: changes topology by connecting or disconnecting equipment.

3.3 Hand calculation

A balanced 33-kV load consumes 20 MW at 0.95 power factor. Estimate its current:

$$I = \frac{P}{\sqrt{3}V\mathrm{PF}} = \frac{20 \times 10^6}{\sqrt{3}(33 \times 10^3)(0.95)} \approx 368 \text{ A}.$$

Repeat the calculation at 110 kV. The comparison makes the benefit of higher transmission voltage tangible.

3.4 PowerFactory practical lab – build the network skeleton

PowerFactory task

Task. Create the course project, create the single-line network topology and learn where the same objects appear in the Data Manager. This week is about connectivity and object handling; detailed line and transformer Types are entered in Week 2.

3.4.1 Task 1 – create the project and base Study Case

Where to click

```
File -> New -> Project
```

1. Set **Project Name** to `PF_12Week_Training` and confirm.
2. If PowerFactory opens a **New Grid** dialog automatically, set the grid name to `TrainingGrid`.
3. Set nominal/system frequency to **50 Hz** when the field is offered.
4. Open the Data Manager. Expand `Study Cases`. Rename the automatically created active Study Case to `W01_BuildNetwork`. If no Study Case was created, right-click the Study Cases folder and create one, then activate it.
5. Save the project with `File -> Save` or `Ctrl+S`.

Why: a project is the database container, the grid contains the electrical network, and the Study Case is the calculation workspace that will later hold Load Flow, Short-Circuit and RMS commands.

3.4.2 Task 2 – place and name the five buses

Open the single-line diagram for `TrainingGrid`. From the Drawing Toolbox choose **Terminal / Busbar** and create five terminals. Double-click each terminal and enter the following on **Basic Data**:

Name	Nominal voltage	Purpose
`Bus220`	220 kV	external-grid connection
`Bus110A`	110 kV	T1 receiving / transmission sending bus
`Bus110B`	110 kV	remote transmission bus
`Bus33A`	33 kV	T2 LV bus
`Bus33B`	33 kV	remote load / future PV bus

After pressing **OK**, verify that the nominal-voltage value shown in each object's dialog is correct. Nominal voltage is model data; it is not the calculated operating voltage.

3.4.3 Task 3 – connect the source, transformers and lines

Use the Drawing Toolbox and create the following objects in this order:

1. **External Grid**: connect it to `Bus220`. Double-click it, set **Name** = `GridSource`. On the **Load Flow** page set the voltage set-point to **1.00 pu**; set reference angle to **0 deg** if this field is shown for the selected mode.
2. **Two-Winding Transformer**: connect `Bus220` to `Bus110A`. Set **Name** = `T1`. Do not invent detailed transformer impedances yet.
3. Create **Line L1** from `Bus110A` to `Bus110B`.
4. Create a second parallel **Line L2** between the same two buses. Place it separately on the diagram so that both circuits are visible.
5. Create **T2** from `Bus110B` to `Bus33A`.
6. Create **Line L33** from `Bus33A` to `Bus33B`.

Why: this creates two parallel 110-kV transfer paths. Later, opening either L1 or L2 will force its power onto the remaining path, giving a useful contingency example.

3.4.4 Task 4 – add the three loads

From the Drawing Toolbox choose **General Load** and connect:

- `LoadA` to `Bus33A`;
- `LoadB` to `Bus33B`;
- `LoadC` to `Bus110B`.

For now enter only the names. The MW and power-factor values are entered in Week 2 after the network equipment is properly typed.

3.4.5 Task 5 – prove that the diagram is a view of database objects

Where to click

`Data Manager -> Network Model -> Network Data -> TrainingGrid`

Locate `Bus220`, `L1`, `T1` and `LoadA`. Select an object in the Data Manager, open its edit dialog, and confirm that it is the same object you see on the SLD. Rename `LoadA` temporarily to `LoadA_TEST` in the Data Manager and verify the name changes on the diagram; then restore `LoadA`.

3.4.6 Task 6 – topology experiment

1. Right-click `L33` on the SLD.
2. Select `Out of Service` if that command is offered in the context menu. Alternatively open L33's edit dialog and tick the **Out of Service** state.
3. Observe the visual state of the element and the fact that `Bus33B/LoadB` no longer has an electrical path to the source.
4. Restore L33 using `In Service` or by clearing the out-of-service state.

Do not judge this week by whether a load flow solves; the line/transformer types are intentionally incomplete until Week 2.

What to verify before moving on

Before finishing, the Data Manager should contain five terminals, three lines, two transformers, three loads and one external grid, all with the names above. Trace a continuous path with your finger from `GridSource` to `LoadB`. Then explain what opening L33 does electrically.

End-of-week checkpoint

Without a tutorial, create a bus, external grid, transformer, line and load; connect them correctly; and explain the difference between the network diagram, the underlying data objects and a Study Case.

Chapter 4

Week 2 – Per-Unit System and Equipment Modelling

Learning outcomes

Understand base quantities, per-unit impedance, line parameters, transformer impedance, ratings, and the distinction between shared equipment Types and individual Elements.

4.1 Three-hour breakdown

0:00–0:50	Theory: per-unit system, line model, transformer model and equipment ratings.
0:50–1:05	Hand calculations: base impedance and conversion between pu and ohms.
1:05–2:30	PowerFactory: create line/transformer types; enter parameters; complete the training network.
2:30–2:50	Experiment: double line length and predict impedance/loading changes.
2:50–3:00	Record where each common datasheet quantity is entered and what physical effect it has.

4.2 Theory: why per-unit is used

Power systems contain many voltage levels. Referring all impedances to one side of every transformer in ohms is cumbersome. The per-unit system normalises quantities relative to selected bases.

Choose a three-phase apparent-power base S_{base} and line-line voltage base V_{base}. Then

$$I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}V_{\text{base}}}, \qquad Z_{\text{base}} = \frac{V_{\text{base}}^2}{S_{\text{base}}}.$$

A per-unit impedance is

$$Z_{\text{pu}} = \frac{Z_{\text{actual}}}{Z_{\text{base}}}.$$

Percentage impedance is simply

$$Z_{\%} = 100 Z_{\text{pu}}.$$

A transformer with approximately 12% short-circuit impedance has $Z_{\text{pu}} \approx 0.12$ on its own rated base.

4.2.1 Changing base

If a per-unit impedance is given on old bases,

$$Z_{\text{pu,new}} = Z_{\text{pu,old}} \left(\frac{S_{\text{new}}}{S_{\text{old}}}\right) \left(\frac{V_{\text{old}}}{V_{\text{new}}}\right)^2.$$

You do not need to perform this conversion every day in PowerFactory, but understanding it prevents incorrect comparison of generator, transformer and network impedances.

4.3 Transmission-line model

A line is represented by series impedance and shunt admittance. Per unit length,

$$z' = R' + jX', \qquad y' = G' + jB'.$$

For length L,

$$Z = (R' + jX')L.$$

A common balanced steady-state representation is the nominal-π model, with the line's shunt admittance divided approximately equally between its two ends.

Physical interpretation:

- R: real-power loss and part of voltage drop.
- X: reactive voltage drop and power-transfer behaviour.
- B: capacitive charging, especially significant on long high-voltage lines/cables.
- Current rating: thermal capability, separate from electrical impedance.

4.4 Transformer model

A transformer model normally includes:

- rated power;
- rated HV/LV voltages;
- winding connection/vector group;
- short-circuit impedance;
- resistive component / copper loss;
- magnetising branch where relevant;
- tap range, tap step and tap side;

- grounding/neutral data for unbalanced and fault studies.

The short-circuit impedance strongly influences both load-flow voltage drop and short-circuit current. A larger transformer impedance generally increases voltage drop under load and reduces downstream fault current.

4.5 Ratings versus operating values

Do not confuse:

- **rated MVA** with actual MVA;
- **nominal voltage** with calculated voltage;
- **rated current** with calculated current;
- **equipment impedance** with system equivalent impedance;
- **thermal limit** with a load-flow control target.

4.6 Type objects and element objects

PowerFactory often separates reusable Type information from individual element information.

A useful mental model is:

Type: what the equipment is	Element: where/how this particular item is used

For example, ten lines may use the same conductor/line type while having different lengths, names, locations and service states. This separation reduces duplicated data and makes fleet-wide updates possible.

4.7 Hand calculation

On $S_{\text{base}} = 100$ MVA and $V_{\text{base}} = 110$ kV,

$$Z_{\text{base}} = \frac{110^2}{100} = 121\ \Omega.$$

Therefore 0.10 pu corresponds to

$$Z = 0.10(121) = 12.1\ \Omega.$$

For 33 kV on the same 100-MVA base,

$$Z_{\text{base}} = \frac{33^2}{100} = 10.89\ \Omega.$$

Notice how the ohmic base changes with the square of voltage.

4.8 PowerFactory practical lab – enter engineering data and run the first sanity load flow

PowerFactory task

Task. Convert the training equipment data into PowerFactory Types and element data. At the end of the lab the network must have enough credible steady-state data to solve its first load flow.

4.8.1 Task 1 – create the 110-kV line Type

Double-click `L1`. On **Basic Data**, find the **Type** field and create a **New Project Type / Line Type** (class `TypLne`). Name it `OH_110kV_Training`.

Enter the following Type values using the closest fields in the line-Type dialog:

Parameter	Value
Rated / nominal voltage	110 kV
Rated current	1.00 kA
Positive-sequence resistance R'	0.08 Ω/km
Positive-sequence reactance X'	0.35 Ω/km
Positive-sequence shunt susceptance B'	3.2 μS/km
Frequency	50 Hz, if requested

If your Type dialog asks for capacitance rather than susceptance, use approximately $C' \approx$ 10.2 nF/km at 50 Hz, because $B' = 2\pi f C'$. Enter *either* the representation requested by the dialog—do not independently add both as if they were separate shunt quantities.

Back in `L1` **Basic Data**, set **Length = 40 km**.

Open `L2`, assign the same `OH_110kV_Training` Type and set **Length = 55 km**.

What you are learning: the Type stores the conductor/circuit electrical construction; the line element stores where that type is used and its installed length.

4.8.2 Task 2 – create the 33-kV line Type

Open `L33` and create a second Line Type named `OH_33kV_Training`. Enter:

Rated voltage	33 kV
Rated current	0.60 kA
R'	0.15 Ω/km
X'	0.32 Ω/km
B'	3.0 μS/km
Length on L33 element	20 km

4.8.3 Task 3 – create the T1 transformer Type

Double-click `T1`. On **Basic Data**, create a new project **Two-Winding Transformer Type** (`TypTr2`) named `TR2_250MVA_220_110`.

Enter:

Rated power	250 MVA
Rated HV voltage	220 kV
Rated LV voltage	110 kV
Short-circuit voltage u_k	12%
Resistive short-circuit component u_{kr}	0.5%
Vector group	YNyn0 for this educational model
Frequency	50 Hz

If your dialog requests rated copper loss rather than u_{kr}, use the field definition shown by PowerFactory and enter a value consistent with about 0.5% resistive impedance on rated current; for this training transformer, approximately 1.25 MW is a convenient educational value. Do not enter both representations unless the model explicitly expects both.

If the Type includes tap-changer data, prepare it now for Week 5:

- tap side: HV;
- neutral tap: 0;
- minimum tap: -10;
- maximum tap: +10;
- voltage step per tap: 1.25%.

4.8.4 Task 4 – create the T2 transformer Type

Repeat for `T2`, creating `TR2_100MVA_110_33`:

Rated power	100 MVA
Rated HV/LV voltages	110 / 33 kV
u_k	10%
u_{kr}	0.5%
Vector group	YNyn0, educational model
Tap side	HV
Neutral / min / max tap	0 / -8 / +8
Tap step	1.25%

4.8.5 Task 5 – enter the load operating data

Open each General Load. Go to the **Load Flow** tab. Set the load to **Balanced** operation. Select the input mode **P, cos(phi) / P, PF** (wording varies slightly). Enter:

Load	Active power	Power factor	Character
`LoadA`	25 MW	0.95	lagging / inductive
`LoadB`	15 MW	0.92	lagging / inductive
`LoadC`	20 MW	0.97	lagging / inductive

If the load dialog uses a sign/inductive-capacitive selector, choose the setting that makes these loads *consume* reactive power. After solving the load flow, the sign of Q in the result will confirm your convention.

4.8.6 Task 6 – first sanity load flow

Where to click

`Main Toolbox / Calculation -> Calculate Load Flow`

Open **Calculate Load Flow** (command class `ComLdf`). On **Basic Options**:

- use the balanced / positive-sequence network representation for this course;
- keep the standard Newton–Raphson load-flow method/default solver unless you intentionally want to compare methods;
- leave advanced controls at their default settings this week.

Click **Execute**. Read the **Output Window**. The calculation should finish successfully. If not, correct errors such as missing Type data, wrong voltage levels, disconnected terminals or equipment out of service.

4.8.7 Task 7 – line-length experiment

1. With the successful base case, record L1 and L2 loading and `Bus110B` voltage.
2. Open `L1 -> Basic Data`. Change **Length** from 40 km to **80 km**. Do not change the Line Type.
3. Execute the same Load Flow command again.
4. Record L1 loading, L2 loading, Bus110B voltage and total/branch losses.
5. Restore L1 to **40 km** and execute Load Flow once more so the project finishes the week in its baseline state.

What to verify before moving on

Before moving on: both 110-kV lines must share one Type but have different lengths; each transformer must have its own Type; all loads must have MW/PF data; and the baseline load flow must execute without a model-data error. Explain why changing L1 length changes network impedance without changing the conductor Type.

End-of-week checkpoint

Given a transformer or line datasheet, identify which quantities describe the equipment itself, which describe how it is installed/operated, and what effect its impedance and rating have on steady-state and fault behaviour.

Chapter 5

Week 3 – Active and Reactive Power, Voltage Drop, Losses and Load Flow

Learning outcomes

Understand what load flow calculates; interpret P, Q, voltage, current, losses and loading; and use PowerFactory results to trace power through a network.

5.1 Three-hour breakdown

0:00–0:50	Theory: complex power, power factor, line losses, voltage-drop intuition and load-flow purpose.
0:50–1:05	Hand calculation: convert MW and PF into MVAr/MVA/current.
1:05–2:30	PowerFactory: run load flow; display bus, branch and transformer results.
2:30–2:50	Experiment: progressively increase one load and predict each result trend.
2:50–3:00	Summarise where power comes from, where it goes and what limits are approached.

5.2 Complex power and power factor

Complex power is

$$S = P + jQ.$$

For a load with known active power and power factor,

$$|S| = \frac{P}{\mathrm{PF}},$$

and for lagging power factor,

$$Q = P \tan(\cos^{-1} \mathrm{PF}).$$

Example: 20 MW at 0.95 lagging:

$$Q = 20 \tan(\cos^{-1} 0.95) \approx 6.57 \text{ MVAr},$$

$$|S| \approx 21.05 \text{ MVA}.$$

Reactive demand increases current even though active MW is unchanged.

5.3 Why voltage changes with loading

For a simplified short AC branch, an engineering approximation is

$$\Delta V \approx \frac{RP + XQ}{V}.$$

This is not a full AC load-flow equation, but it builds correct intuition:

- greater P usually increases voltage drop;
- greater inductive Q usually increases voltage drop;
- large X/R networks are particularly sensitive to reactive-power transfer;
- local reactive support can reduce reactive current transported over the network.

5.4 Losses

For a balanced line,

$$P_{\text{loss}} = 3I^2R.$$

If current doubles, resistive loss approximately quadruples. This explains why heavily loaded networks may experience rapidly increasing losses.

5.5 What load flow solves

At every bus, AC load flow seeks a consistent set of:

$$V_i, \quad \delta_i, \quad P_i, \quad Q_i$$

subject to network admittance relationships, generator controls, load demands and limits.

At a conceptual level, Kirchhoff's current law is enforced using complex voltages and admittances. For bus i,

$$I_i = \sum_{k=1}^{n} Y_{ik} V_k,$$

and complex power injection is

$$S_i = V_i I_i^*.$$

Substituting the network currents gives nonlinear power-balance equations. That is why iterative numerical methods are required.

5.6 PowerFactory practical lab – run, display and diagnose a load flow

PowerFactory task

Task. Run a reproducible load flow, configure the result display, trace MW/MVAr through the network and perform a four-point load-growth experiment.

5.6.1 Task 1 – save the normal operating condition

With the Week 2 values active, create an Operation Scenario named `OP_Normal` using `File -> Save Operation Scenario as...`. Save it. This gives you a known state to return to after experiments.

5.6.2 Task 2 – configure and execute Load Flow

Open `Calculate Load Flow` (`ComLdf`). On **Basic Options** use:

- network representation: balanced / positive sequence;
- load-flow method: standard Newton–Raphson/default method;
- active-power balance: external/reference source balances the remaining system power;
- reactive-power limits: may be enabled; it becomes important after G1 is added in Week 4.

Press **Execute**. In the Output Window look for successful completion and check that there are no warnings indicating isolated equipment, invalid data or controller conflicts.

5.6.3 Task 3 – show the results on the SLD

After a successful calculation, PowerFactory normally displays result boxes/labels on the single-line diagram. If the labels are hidden, use the diagram/results display controls to show calculation results.

For each terminal, display or inspect:

- line-line voltage in kV;
- voltage in pu;
- voltage angle in degrees.

For each line and transformer, display or inspect:

- active power P at each end;
- reactive power Q at each end;
- current;
- loading in percent.

If the SLD is crowded, open the **Network Model Manager** and use its **Flexible Data** page/table to inspect the same calculated variables.

5.6.4 Task 4 – complete the baseline result sheet

Record these values before changing anything:

Quantity	Record
Bus110A, Bus110B, Bus33A, Bus33B	voltage pu
L1, L2, L33	loading % and P/Q direction
T1, T2	loading % and P/Q
GridSource	MW and MVAr supplied
LoadA/B/C	MW and MVAr consumed
Network	approximate active losses

To estimate a branch active-power loss from results, compare the active power entering and leaving that branch using a consistent sign convention.

5.6.5 Task 5 – four-point load-growth experiment

Activate `OP_Normal`. Open `LoadA -> Load Flow`. Keep PF at 0.95 and run the following cases:

LoadA P (MW)	Bus33A V (pu)	T2 loading (%)	GridSource P (MW)	GridSource Q (MVAr)
25				
40				
60				
80				

For each row:

1. enter the new **Active Power** value in LoadA;
2. click **OK**;
3. execute the same `ComLdf`;
4. confirm convergence in the Output Window;
5. record Bus33A voltage, T2 loading, GridSource P/Q and, if convenient, losses;
6. before looking at the result, write arrows ↑ or ↓ for what you expected.

When finished, reactivate `OP_Normal` or restore LoadA to 25 MW / 0.95 PF and save the normal scenario.

5.6.6 Why this lab matters

You are practising the full study loop:

$$\text{define operating point} \rightarrow \text{execute calculation} \rightarrow \text{inspect output}$$
$$\rightarrow \text{extract results} \rightarrow \text{explain physics.}$$

What to verify before moving on

You should be able to point to a branch and state which direction active power flows, identify the lowest-voltage bus and highest-loading element, and explain why the 80-MW case has more current/loss and usually lower remote voltage than the 25-MW case.

End-of-week checkpoint

Given a converged load-flow result, explain the direction of MW and MVAr flow, identify the weakest bus, identify the most heavily loaded element, and explain why losses changed when load was increased.

Chapter 6

Week 4 – Load-Flow Bus Types, Slack-/PV/PQ Control and Convergence

Learning outcomes

Understand PQ, PV and slack/reference buses, the role of the external grid, generator voltage control, and the numerical idea behind Newton–Raphson load flow.

6.1 Three-hour breakdown

0:00–0:55	Theory: P, Q, V, δ, PQ/PV/slack buses, nonlinear equations and Newton–Raphson concept.
0:55–1:10	Hand reasoning: identify specified and solved variables for three example buses.
1:10–2:30	PowerFactory: configure external grid and voltage-controlled generator; create light/normal/peak cases.
2:30–2:50	Experiment: vary load power factor at constant MW.
2:50–3:00	Record how generator MVAr and voltage change as demand becomes more inductive.

6.2 The four bus quantities

For balanced AC load flow the important quantities are

$$P, \quad Q, \quad |V|, \quad \delta.$$

Only two are independently specified at each conventional bus; the other two are solved.

6.2.1 PQ bus

At a PQ bus,

$$P,\ Q \quad \text{specified}, \qquad |V|,\ \delta \quad \text{solved}.$$

A conventional constant-power load bus is the common example.

6.2.2 PV bus

At a PV bus,

$$P,\ |V| \quad \text{specified}, \qquad Q,\ \delta \quad \text{solved}.$$

A synchronous generator controlling terminal voltage is the typical example. Reactive power is adjusted as needed, subject to $Q_{\min}$ and $Q_{\max}$.

6.2.3 Slack/reference bus

At the slack bus,

$$|V|,\ \delta \quad \text{specified}, \qquad P,\ Q \quad \text{solved}.$$

Why is this needed? Total specified generation and load will not perfectly account for losses before the solution is known. The slack source absorbs the mismatch and sets the angular reference, usually $\delta = 0$.

6.3 Newton–Raphson intuition

The AC power-flow equations are nonlinear. Define a mismatch vector between specified and calculated power:

$$\Delta\mathbf{f} = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix}.$$

Newton–Raphson linearises the equations around the current estimate:

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \mathbf{J} \begin{bmatrix} \Delta\delta \\ \Delta V \end{bmatrix},$$

where $\mathbf{J}$ is the Jacobian. The algorithm repeatedly:

1. assumes a voltage/angle state;
2. calculates power injections;
3. computes mismatch;
4. solves for corrections;
5. updates the state;
6. stops when mismatch is sufficiently small.

You do not need to derive every Jacobian term at this stage. You do need to understand that "non-convergence" often indicates bad data, impossible controls, isolated islands, severe operating stress or poor numerical conditions—not simply "software failure".

6.4 PowerFactory practical lab – create a PV generator bus and operating scenarios

PowerFactory task

Task. Add a synchronous generator that controls its terminal voltage, keep the External Grid as the slack/reference source, and create light/normal/peak operating scenarios. Then vary load power factor at fixed MW.

6.4.1 Task 1 – place and rate synchronous generator G1

From the Drawing Toolbox choose **Synchronous Machine** and connect it to `Bus110B`. Open the machine dialog:

1. **Basic Data**: Name = `G1`; operating category = Generator if a generator/motor selector is present.
2. **Type**: create or assign a project Synchronous Machine Type named `GEN_50MVA_110kV`. Enter rated apparent power **50 MVA**, rated voltage **110 kV** and rated PF approximately **0.85**. Detailed transient constants are deliberately deferred to Week 9.

6.4.2 Task 2 – make G1 a voltage-controlled PV source

Open `G1 -> Load Flow`. Configure the local controller / operating mode so that the generator specifies **active power and voltage**. Depending on your version this may appear as **Mode of Local Voltage Controller = Voltage**, or as an input/control mode exposing P and U.

Enter:

Active power set-point	25 MW
Voltage set-point	1.00 pu
Minimum reactive power	-20 MVAr
Maximum reactive power	+20 MVAr
Reference/slack machine	disabled for G1

If reactive limits are stored in the machine Type rather than the element in your selected model, put the same educational limits there or choose the element setting that permits local limits.

6.4.3 Task 3 – confirm the External Grid is the slack/reference

Open `GridSource -> Load Flow`. Confirm voltage = **1.00 pu** and reference angle = **0 deg** where the field is available. The External Grid remains the reference; do not make G1 a second reference machine for this exercise.

6.4.4 Task 4 – execute load flow with Q limits

Open `Calculate Load Flow -> Basic Options`. Enable **Consider Reactive Power Limits** (or equivalent wording) so that voltage-controlled generators cannot supply unlimited MVAr. Execute.

Verify:

- G1 active power is approximately 25 MW;
- G1 reactive power is a calculated result, not the fixed input;

- Bus110B is held near the 1.00-pu target while G1 remains inside its Q limits;
- GridSource supplies the remaining active power plus network losses.

6.4.5 Task 5 – create three Operation Scenarios

Create the following through `File -> Save Operation Scenario as...` (or the Operation Scenarios folder in the Data Manager). Make sure you **save** each scenario after changing the load values.

Scenario	LoadA	LoadB	LoadC
`OP_LightLoad`	15 MW @ 0.95	9 MW @ 0.92	12 MW @ 0.97
`OP_Normal`	25 MW @ 0.95	15 MW @ 0.92	20 MW @ 0.97
`OP_PeakLoad`	35 MW @ 0.95	21 MW @ 0.92	28 MW @ 0.97

For each scenario: right-click it and select **Activate**; execute Load Flow; record minimum bus voltage, maximum branch loading, G1 Q and GridSource P/Q.

6.4.6 Task 6 – power-factor experiment at fixed MW

Activate `OP_Normal`. Temporarily set `LoadA` active power to **40 MW**. Then perform:

$$\mathrm{PF} = 0.98, agCase1$$

$$\mathrm{PF} = 0.95, agCase2$$

$$\mathrm{PF} = 0.90.agCase3$$

For each case open `LoadA -> Load Flow`, change only the PF, execute `ComLdf`, and record:

- LoadA MVAr;
- Bus33A and Bus33B voltage;
- T2 current/loading;
- G1 MVAr;
- GridSource MVAr.

At the end, reactivate the saved `OP_Normal` scenario so the original 25-MW LoadA is restored.

What to verify before moving on

Open G1 after the calculation. You should be able to distinguish its *specified* variables (P and voltage) from its *solved* reactive-power result. Explain why G1 Q changes when the load PF changes even though G1 P remains at 25 MW.

End-of-week checkpoint

Explain why two 40-MW loads with different power factors can create different currents and voltage conditions, and explain why the slack bus and PV generator do not have the same specified variables.

Chapter 7

Week 5 – Voltage Control, Reactive Power, Generator Limits and Transformer Taps

Learning outcomes

Diagnose low/high voltage conditions and compare generator reactive support, transformer tap control and shunt compensation.

7.1 Three-hour breakdown

0:00–0:50	Theory: P-frequency and Q-voltage intuition, reactive sources, Q limits, taps and voltage control.
0:50–1:05	Hand exercise: compute MVAr of a load and estimate effect of local compensation.
1:05–2:35	PowerFactory: intentionally create a low-voltage case and solve it three different ways.
2:35–2:50	Experiment: force generator to a Q limit and observe loss of voltage control.
2:50–3:00	Write a diagnosis tree for a low-voltage bus.

7.2 Why reactive power is closely linked to voltage

A useful first-order rule is:

$$\text{active-power imbalance} \leftrightarrow \text{frequency}, \qquad \text{reactive-power balance} \leftrightarrow \text{voltage}.$$

The coupling is not perfectly independent, but the rule is extremely useful. In many transmission systems where $X \gg R$, reactive-power transfer has a strong effect on voltage magnitude.

7.3 Sources and sinks of reactive power

- Inductive motors and transformers generally consume reactive power.
- Shunt capacitors inject capacitive reactive power.

- Shunt reactors absorb reactive power.
- Synchronous generators can inject or absorb reactive power within capability limits.
- Inverter-based resources can provide reactive support within converter/current and grid-code constraints.

7.4 Generator reactive limits

A generator cannot provide unlimited MVAr. A simplified constraint is

$$Q_{\min} \leq Q_G \leq Q_{\max}.$$

A voltage-controlled generator behaves like a PV bus only while the required reactive output remains feasible. If it hits a reactive limit, it can no longer freely adjust Q to maintain the voltage target. The practical consequence can be a voltage drift away from set point.

7.5 Transformer tap control

Changing a transformer tap changes its effective turns ratio. An on-load tap changer can adjust voltage without de-energising the transformer.

Important concepts:

- controlled bus;
- target voltage;
- deadband;
- tap step;
- minimum/maximum tap;
- control side and sign convention.

A deadband prevents excessive tap operations for tiny voltage deviations.

7.6 Shunt compensation

Suppose a load consumes

$$P = 40 \text{ MW}, \qquad \text{PF} = 0.90.$$

Then

$$Q_{\text{load}} = 40 \tan(\cos^{-1} 0.90) \approx 19.37 \text{ MVAr}.$$

If a local capacitor supplies 10 MVAr, upstream reactive demand becomes approximately 9.37 MVAr, neglecting network changes. The active load remains 40 MW.

This can reduce upstream current and voltage drop.

7.7 PowerFactory practical lab – deliberately create and solve a voltage problem

PowerFactory task

Task. Create a low-voltage condition and correct it using three different mechanisms: generator reactive control, transformer tap change and local shunt compensation. Then deliberately hit the generator Q limit.

7.7.1 Task 1 – create a voltage-stress Operation Scenario

Activate `OP_Normal`. Then choose

```
File -> Save Operation Scenario as...
```

and create `OP_VoltageStress`. In this new scenario set:

Load	P	PF
LoadA	60 MW	0.90 lagging
LoadB	30 MW	0.90 lagging
LoadC	20 MW	0.97 lagging

Set G1 voltage target to 1.00 pu and Q limits to $-20, +20$ MVAr. Execute Load Flow with reactive limits enabled. Record Bus33A/Bus33B voltage. If Bus33B is still above approximately 0.95 pu, increase LoadA or LoadB in 5–10 MW increments until you obtain a clear but convergent low-voltage condition near 0.92–0.95 pu. **Write down the exact load values that created your starting condition**; use those same values for all three remedies.

7.7.2 Task 2 – Remedy A: generator voltage/reactive support

1. Open `G1 -> Load Flow`.
2. Confirm the local mode is **Voltage control** / P-U mode.
3. Run once with **Uset = 1.00 pu**; record G1 Q and Bus33B voltage.
4. Change **Uset = 1.02 pu**.
5. Execute Load Flow again.
6. Record G1 Q, Bus110B voltage, Bus33B voltage, T2 Q flow and T2 loading.

If G1 immediately reaches $Q_{\max}$, that is a useful result: voltage control is constrained by machine capability.

7.7.3 Task 3 – Remedy B: transformer T2 tap

First restore G1 voltage set-point to 1.00 pu and execute Load Flow to re-establish the stressed baseline.

Open `T2`. Find **Tap Position** on the element's Basic Data / Load Flow / operating-data page (the exact tab depends on model/version).

1. Record the neutral/base tap, normally 0 in this project.
2. Change the tap by **one step** (+1) and execute Load Flow.
3. Observe Bus33B voltage.

4. If the voltage moved in the wrong direction, return to 0 and try -1 instead. Tap sign depends on the defined tap side and transformer ratio convention; never assume the sign without checking the resulting ratio/voltage.
5. Continue one step at a time until you can demonstrate a meaningful downstream-voltage change. Stay within the configured tap limits.
6. Record tap position, Bus33A, Bus33B, T2 P/Q and T2 loading.

Return T2 to neutral tap after the comparison unless you explicitly save a separate scenario.

7.7.4 Task 4 – Remedy C: local capacitor at Bus33B

From the Drawing Toolbox select **Shunt / Shunt Capacitor** (`ElmShnt`) and connect it to `Bus33B`. Open its dialog:

- Name = `C_Bus33B`;
- nominal voltage = 33 kV where requested;
- shunt type = capacitive / capacitor;
- first case reactive rating = approximately 5 MVAr at nominal voltage (or configure one 5-MVAr step);
- in service = yes.

Execute Load Flow and record Bus33B voltage, T2 MVAr flow and T2 loading/current. Then change the connected compensation to approximately **10 MVAr** (e.g. two 5-MVAr steps or a 10-MVAr rating according to the chosen model) and repeat.

Check the sign of the calculated shunt Q so you know that it is supplying capacitive reactive power under your result convention.

7.7.5 Task 5 – force G1 to its reactive-power limit

Return T2 to neutral and place the capacitor out of service so that only G1 is attempting to regulate voltage.

1. Open `G1 -> Load Flow`.
2. Set $Q_{\text{max}} = +5$ MVAr and $Q_{\text{min}} = -5$ MVAr for this experiment.
3. Set voltage target to **1.02 pu**.
4. Confirm **Consider Reactive Power Limits** is enabled in `ComLdf`.
5. Execute Load Flow.
6. If G1 has not reached +5 MVAr, worsen LoadA/LoadB power factor gradually (for example $0.90 \rightarrow 0.87 \rightarrow 0.85$) while keeping MW fixed and rerun.
7. Stop once the generator Q is clamped at/near the limit. Record the actual controlled-bus voltage and Bus33B voltage.

Restore G1 limits to $-20, +20$ MVAr afterward and save the intended scenario state.

7.7.6 Result comparison to write in your notebook

Method	Control changed	Bus33B V	G1 Q	T2 Q / loading
Baseline	none			
Generator	G1 Uset 1.02 pu			
Tap	T2 one/more steps			
Capacitor	+5 / +10 MVAr			
Q-limit case	G1 Qmax = 5 MVAr			

What to verify before moving on

You should be able to show that a tap change, local MVAr injection and generator excitation do not solve voltage problems in exactly the same way. Also demonstrate one case where G1 reaches its Q limit and therefore cannot hold its requested voltage perfectly.

End-of-week checkpoint

When presented with a 0.91-pu bus, ask: Is reactive demand excessive? Is the source weak? Is the generator at a Q limit? Are taps appropriate? Would local compensation reduce reactive transfer? Do not jump directly to a voltage set-point change.

Chapter 8

Week 6 – Short-Circuit Fundamentals and Symmetrical Components

Learning outcomes

Understand Thevenin equivalent impedance, fault level, source strength, sequence components, major fault types and the purpose of standardised short-circuit calculations.

8.1 Three-hour breakdown

0:00–1:00	Theory: Thevenin fault model, source strength, generator subtransient response and sequence networks.
1:00–1:15	Hand calculation: three-phase fault from equivalent impedance.
1:15–2:30	PowerFactory: configure source fault strength and calculate faults at 220, 110 and 33 kV.
2:30–2:50	Experiment: increase source impedance / reduce short-circuit power and compare results.
2:50–3:00	Explain in words why the fault level changed.

8.2 Thevenin view of a fault

Any linear network seen from a fault location can conceptually be replaced by a Thevenin voltage source behind an equivalent impedance:

$$I_f \approx \frac{V_{\text{th}}}{Z_{\text{th}} + Z_f},$$

where Z_f is fault impedance. For a bolted fault $Z_f \approx 0$.

This immediately gives the most important fault-level intuition:

$$|Z_{\text{th}}| \downarrow \quad \Rightarrow \quad |I_f| \uparrow .$$

A "strong" system has relatively low source impedance and high short-circuit power. A "weak" system has higher equivalent impedance and lower fault level.

8.3 Short-circuit power

For a balanced three-phase fault, an approximate relationship is

$$S''_k = \sqrt{3}\,V_n I''_k.$$

Hence, if an external grid is represented by S''_k,

$$I''_k \approx \frac{S''_k}{\sqrt{3}V_n}.$$

This is why an external-grid short-circuit power is a convenient way to describe system strength.

8.4 Generator contribution

Immediately after a fault, a synchronous generator's current is strongly influenced by its subtransient reactance X''_d. In simplified form,

$$I'' \sim \frac{E''}{X''_d}.$$

The generator contribution then evolves as electromagnetic transients decay. For equipment-duty calculations, the initial/subtransient current is particularly important.

8.5 Symmetrical components

An unbalanced set of three-phase phasors can be decomposed into:

- positive-sequence components;
- negative-sequence components;
- zero-sequence components.

8.5.1 Positive sequence

Three equal magnitudes separated by 120° in the normal phase order. Positive sequence represents the balanced operating system and is the only sequence present in an ideal balanced three-phase condition.

8.5.2 Negative sequence

Three equal magnitudes separated by 120° in the reverse phase order. Negative-sequence currents can cause additional heating in rotating machines.

8.5.3 Zero sequence

Three components equal in magnitude and angle. Zero-sequence current needs a return path, normally involving neutral/earth connections. Consequently, transformer vector groups and grounding arrangements can radically change ground-fault current.

8.6 Fault types

Three-phase (LLL): balanced; primarily positive-sequence.

Single-line-to-ground (LG): unbalanced; involves positive, negative and zero sequences.

Line-to-line (LL): unbalanced; no zero-sequence current for a simple LL fault.

Double-line-to-ground (LLG): unbalanced; generally involves all three sequence networks.

8.7 Standardised calculations

IEC 60909 provides a standardised method for short-circuit current calculation. The exact standard contains correction factors and definitions beyond the scope of this first course. The essential lesson is that fault studies used for equipment specification should follow an agreed calculation method rather than an ad-hoc pre-fault assumption.

8.8 Hand calculation

For

$$V_{LL} = 33 \text{ kV}, \qquad Z_{\text{th}} = j3\ \Omega,$$

phase voltage is

$$V_{\text{ph}} = \frac{33}{\sqrt{3}} = 19.05 \text{ kV}.$$

Thus

$$I_f \approx \frac{19.05}{3} = 6.35 \text{ kA}.$$

8.9 PowerFactory practical lab – configure IEC source strength and calculate three-phase faults

PowerFactory task

Task. Enter the external-grid short-circuit strength, run IEC 60909 three-phase faults at several voltage levels and then deliberately weaken the source to observe the effect on I_k''.

8.9.1 Task 1 – enter External Grid short-circuit data

Open `GridSource`. Go to the **VDE/IEC Short-Circuit** / **Short-Circuit** page. Enter the educational values:

Maximum initial short-circuit power S_k''	5000 MVA
Minimum initial short-circuit power S_k''	3000 MVA
R/X for maximum case	0.10
R/X for minimum case	0.10

If your field is labelled **X/R** instead of R/X, enter the reciprocal, approximately **10**. Read the field label carefully; mixing these ratios changes the asymmetry/peak-current calculation.

8.9.2 Task 2 – prepare the network

Make sure L1, L2, L33, T1 and T2 are in service. Activate `OP_Normal`. A pre-fault load flow is useful for checking topology even though IEC 60909 uses its own standardised voltage-factor treatment.

For the first three-phase study, the positive-sequence network is dominant, so detailed zero-sequence data is not yet the focus. That will be added in Week 7.

8.9.3 Task 3 – calculate a three-phase fault at Bus220

You can start the command either by right-clicking the target bus and choosing `Calculate -> Short-Circuit`, or by opening **Calculate Short-Circuit** from the main toolbox (keyboard shortcut `Ctrl+F11` in current PowerFactory documentation).

Set:

- **Method**: According to IEC 60909;
- **Fault Type**: 3-Phase Short-Circuit / 3-phase fault;
- **Fault Location**: User Selection = `Bus220`;
- fault resistance/reactance: 0 / 0 if these are exposed for the selected calculation;
- calculation case: maximum initial short-circuit current for the first run.

Click **Execute**. Confirm success in the Output Window.

Record from the bus result box or Network Model Manager:

- I_k'' in kA;
- S_k'' in MVA;
- peak current i_p, if calculated/displayed.

8.9.4 Task 4 – repeat at lower voltage buses

Repeat exactly the same IEC 60909 / 3-phase calculation for:

1. `Bus110A`;
2. `Bus110B`;
3. `Bus33A`;
4. `Bus33B`.

Use a table:

Bus	I_k'' (kA)	S_k'' (MVA)	Main impedance between source and fault
Bus220			source equivalent
Bus110A			T1 + source
Bus110B			T1 + 110-kV network + source
Bus33A			T2 + upstream network
Bus33B			L33 + T2 + upstream network

8.9.5 Task 5 – source-strength experiment

Open `GridSource -> VDE/IEC Short-Circuit`. Change maximum S_k'' from 5000 MVA to **2500 MVA**. Do not change the physical transmission-line/transformer data.

Recalculate the 3-phase fault at `Bus220` and `Bus33B`. Record the new I_k''. Then restore maximum S_k'' to **5000 MVA**.

Interpretation: reducing short-circuit power makes the external system electrically weaker (larger equivalent source impedance), so fault current should decrease.

What to verify before moving on

Show one screenshot/result table containing I_k'' at at least three buses and explain the trend using the impedances between GridSource and the fault. Then show numerically that the weaker 2500-MVA source produces lower fault current than the 5000-MVA source.

End-of-week checkpoint

Explain why fault current is not a fixed property of a nominal voltage level. It is determined by the sources and impedances electrically visible from the fault location, together with the fault type and grounding paths.

Chapter 9

Week 7 – Practical Short-Circuit Studies and Equipment Duty

Learning outcomes

Compare balanced/unbalanced faults, maximum/minimum fault conditions, source contributions, peak current and the significance of grounding and transformer vector groups.

9.1 Three-hour breakdown

0:00–0:45	Theory: initial symmetrical current, peak current, thermal duty, max/min fault level and protection implications.
0:45–1:00	Reasoning exercise: predict effects of adding/removing sources.
1:00–2:35	PowerFactory: LLL, LG, LL and LLG faults; strong/weak grid; generator in/out; parallel transformer tests.
2:35–2:50	Build a fault-level comparison table.
2:50–3:00	Write engineering conclusions rather than only recording currents.

9.2 Important short-circuit quantities

9.2.1 Initial symmetrical short-circuit current I_k''

This is the RMS value of the symmetrical AC component at the beginning of the fault under the selected standard/model assumptions. It is a central quantity in equipment-duty calculations.

9.2.2 Peak current i_p

The total fault current contains an AC component plus a decaying DC offset whose magnitude depends on fault inception and network R/X. The peak current is important for mechanical/-electrodynamic forces.

A conceptual relationship is

$$i_p = \kappa\sqrt{2}I_k'',$$

where κ accounts for asymmetry/DC offset according to the adopted standard and impedance ratio.

9.2.3 Thermal duty

Conductors and switchgear heat during a fault. A simplified thermal concept is related to

$$I^2t.$$

The actual standardised equivalent thermal current calculation considers the fault-current decay and fault duration. The engineering question is whether equipment can withstand the energy until protection clears the fault.

9.3 Maximum and minimum fault currents

Maximum fault current is relevant to:

- breaker interrupting rating;
- making current;
- busbar mechanical withstand;
- cable/transformer thermal stress.

Minimum fault current is relevant to:

- relay pickup sensitivity;
- fuse operation;
- ensuring remote/end-of-feeder faults are detected.

Maximum and minimum system configurations may differ. Maximum generation and parallel circuits often increase fault level; outages and weak-source conditions may reduce it.

9.4 Why ground faults are special

For a single-line-to-ground fault, sequence networks are connected such that zero-sequence impedance matters. Zero sequence is heavily influenced by:

- transformer winding connection;
- neutral grounding;
- grounding transformers;
- earth-return path;
- cable/line zero-sequence parameters.

A delta winding can block zero-sequence current transfer to the other side while still allowing circulating zero-sequence components within the delta. This is why vector group cannot be ignored in earth-fault studies.

9.5 PowerFactory practical lab – unbalanced faults, source contributions and fault duty

PowerFactory task

Task. Extend the model for zero-sequence studies, calculate LLL/LG/LL/LLG faults, compare maximum/minimum conditions and demonstrate how generator, transformer and grounding choices change fault current.

9.5.1 Task 1 – add educational zero-sequence line data

Unbalanced ground faults require zero-sequence data. Open `OH_110kV_Training` and find the **Short-Circuit / Zero Sequence** fields. For this educational model enter, where the Type permits direct input:

Type	R_0'	X_0'
OH_110kV_Training	0.24 Ω/km	1.05 Ω/km
OH_33kV_Training	0.45 Ω/km	0.96 Ω/km

If the selected line model asks for other zero-sequence quantities (e.g. susceptance/capacitance or earth-return parameters), leave nonessential educational fields at documented defaults rather than fabricating project-specific data.

Important modelling note

The zero-sequence values above are deliberately illustrative, not utility or conductor-design data. Real earth-fault studies require validated zero-sequence line/cable impedances, transformer connections and grounding data.

9.5.2 Task 2 – verify transformer vector group and grounding

Open T1 and T2 Types. For this course both are YNyn0 so that grounded star points can provide a zero-sequence path. In the transformer/neutral/earthing fields, ensure the intended neutrals are grounded according to the selected model. If a neutral impedance field is available, use solid grounding for the first comparison (approximately zero neutral impedance).

9.5.3 Task 3 – four fault types at Bus33B

Right-click `Bus33B -> Calculate -> Short-Circuit`. Use **IEC 60909**. Execute the following one at a time, keeping a bolted/zero-impedance fault unless you deliberately study fault resistance:

1. 3-phase short circuit (LLL);
2. single-phase-to-ground (LG);
3. 2-phase short circuit (LL);
4. 2-phase-to-ground (LLG).

After each run record I_k'', S_k'' if applicable and peak current where reported.

9.5.4 Task 4 – maximum versus minimum short-circuit case

In `ComShc`, select the option for the **maximum** short-circuit condition and run LLL and LG at Bus33B. Then select the **minimum** condition and repeat. Record all four values.

The exact maximum/minimum options depend on the selected IEC calculation form, but the calculation command should clearly identify the case/voltage-factor selection.

9.5.5 Task 5 – generator contribution test

Run an LLL fault at `Bus110B` with G1 **in service**. Record I_k''.

Then:

1. right-click `G1 -> Out of Service`;
2. execute the same fault calculation without changing the fault location/method;
3. record I_k'';
4. restore `G1 -> In Service`.

If G1's short-circuit contribution is not yet defined by its Type, PowerFactory may warn or use model defaults. Treat this as a modelling lesson: generator contribution is only credible when its short-circuit parameters (including subtransient data) are credible.

9.5.6 Task 6 – optional parallel-transformer sensitivity

Create a temporary second transformer `T2B` in parallel with T2. Assign the same Type, `TR2_100MVA_110_33`, and connect it between the same 110-kV and 33-kV buses. Ensure both units are in service and calculate an LLL fault at Bus33A. Then set T2B out of service and repeat.

Parallel transformer paths normally reduce equivalent source impedance and increase downstream fault level. Delete T2B or leave it out of service after the experiment so the baseline network remains unchanged.

9.5.7 Task 7 – optional cable/line thermal fault check

If your line Type exposes the field **Rated Short-Time (1s) Current (Conductor)** on the Short-Circuit page, enter an explicitly educational value (for example 20 kA) only to learn the workflow. In the Short-Circuit command set **Fault Clearing Time (Ith)** to **0.20 s**, execute the fault and use diagram colouring/result variables for short-circuit thermal/peak loading if available.

The purpose is to understand how fault magnitude and clearing time are compared with short-time equipment capability—not to establish a real conductor rating.

9.5.8 Task 8 – write the comparison table

Condition	LLL	LG	LL	LLG
Bus33B maximum				
Bus33B minimum				
G1 out (selected fault)				
Parallel T2 (selected fault)				

What to verify before moving on

Your conclusion must contain at least two causal statements, e.g. "removing G1 increased the Thevenin impedance and reduced fault current" and "the LG current is strongly affected by the zero-sequence path through grounded transformer neutrals." Numbers without this explanation do not complete the lab.

End-of-week checkpoint

Explain why adding generation can increase fault level and why an LG fault may be either larger or smaller than an LLL fault depending on sequence impedances and grounding.

Chapter 10

Week 8 – $N-1$ Contingency Analysis and Network Security

Learning outcomes

Understand preventive security, credible outages, thermal/voltage post-contingency limits, power-flow redistribution and the difference between normal and post-fault operating states.

10.1 Three-hour breakdown

0:00–0:45	Theory: $N-1$, outage severity, overloads, voltage violations and redispatch concepts.
0:45–1:00	Base-case screening: select monitored voltages and loadings.
1:00–2:35	PowerFactory: manually test line, transformer and generator outages or use Contingency Analysis if licensed.
2:35–2:50	Rank contingencies by severity.
2:50–3:00	Write one mitigation for the worst case.

10.2 What $N-1$ means

If the intact network has N relevant components, an $N-1$ study considers the loss of one credible component at a time.

The objective is not necessarily that nothing changes. The objective is that the post-contingency system remains within defined security criteria or can be restored acceptably through permitted corrective actions.

Typical monitored criteria are:

- bus voltage range;
- line/transformer loading;
- generator capability;
- islanding / unsupplied load;
- transfer limits.

10.3 Power redistribution

In a meshed network, if one line opens, its previous power does not disappear. It redistributes according to network impedances and injections.

If two parallel paths originally carry power, loss of one path increases loading on the survivor. In a wider mesh the redistribution can be non-intuitive, which is why systematic contingency calculation is needed.

10.4 Security versus reliability

These terms overlap but are not identical.

Security asks whether the system can withstand specified disturbances while remaining in an acceptable operating state.

Reliability is broader and includes probability/frequency/duration of failures and customer interruption over time.

Week 8 focuses on deterministic security.

10.5 PowerFactory practical lab – perform an $N-1$ security assessment

PowerFactory task

Task. Establish a normal base case, remove one component at a time, rerun Load Flow, rank violations and test one corrective action. The manual workflow is mandatory because it teaches the physics; the automated Contingency Analysis workflow is optional if your licence includes it.

10.5.1 Task 1 – create a safe working scenario

Activate `OP_Normal`. Use `File -> Save Operation Scenario as...` to create `OP_N1_Working`. Perform all manual switching in this working scenario so that your saved normal scenario remains untouched.

Run Load Flow with all equipment in service and record:

Base metric	Value
Minimum bus voltage	
Maximum bus voltage	
Highest line loading	
Highest transformer loading	
GridSource MW/MVAr	
Approximate active losses	

10.5.2 Task 2 – manual L1 outage

1. Confirm both L1 and L2 are in service.
2. Right-click `L1 -> Out of Service`.
3. Execute the existing Load Flow command.

4. Read the Output Window. Note whether the calculation converged and whether any island/unsupplied-network warning appeared.
5. Record minimum bus voltage, L2 loading, T1/T2 loading and any limit violations.
6. Trace the changed MW flow on the SLD: the power previously shared by L1 must now use other available paths, principally L2 in this small network.
7. Restore `L1 -> In Service` and rerun the base load flow before the next outage.

10.5.3 Task 3 – repeat one-at-a-time outages

Repeat the exact sequence for:

1. `L2` out;
2. `T2` out;
3. `G1` out.

For T2, this training system may lose the entire 33-kV supply because only one 110/33-kV transformer exists. If the load flow reports an isolated/unsupplied area, that is a correct security finding rather than a software failure.

Use:

Outage	Converged/supplied?	Min V (pu)	Max loading (%)	Main consequence
None				
L1				
L2				
T2				
G1				

10.5.4 Task 4 – optional automated Contingency Analysis

If the Contingency Analysis module is licensed, try one of the following workflows supported by the interface:

1. multi-select L1, L2, T2 and G1, right-click and look under `Calculate -> Contingency Analysis`; or
2. create explicit cases under `Operational Library -> Fault Cases`, using the command to create multiple $n-1$ fault cases from selected components.

Open the **Contingency Analysis** calculation command, select/confirm the created fault cases and execute. Compare its violation list with your manual table. If the module is not present, skip this task; no core learning outcome is lost.

10.5.5 Task 5 – mitigate the worst contingency

Choose the worst *supplied* contingency (for example, one 110-kV line out with the remaining line heavily loaded). Keep that element out of service and perform one controlled mitigation, such as:

- reduce LoadA by 10 MW;
- or change G1 active-power dispatch by 10 MW if the direction of redispatch helps the overloaded corridor;
- or, if voltage rather than loading is the constraint, apply the Week 5 reactive/tap remedy.

Execute Load Flow again and quantify how much the worst loading or voltage improves. Then restore the normal equipment states.

What to verify before moving on

Before finishing, all baseline equipment must be back in service. You should be able to rank the outages and explain the worst case by showing where power was rerouted. The automated module, if used, should confirm rather than replace this understanding.

End-of-week checkpoint

Take the worst contingency and explain the physics: which path was removed, where MW was rerouted, which element became limiting, and what plausible mitigation could relieve it.

Chapter 11

Week 9 – Dynamic Models, Inertia, Frequency and Rotor-Angle Fundamentals

Learning outcomes

Transition from steady-state to time-domain thinking; understand inertia, mechanical/-electrical power balance, rotor angle, frequency, dynamic initialization, events and result channels.

11.1 Three-hour breakdown

0:00–1:00	Theory: synchronous-machine energy balance, swing equation, inertia, stability categories and dynamic controls.
1:00–1:15	Reasoning exercise: predict rotor acceleration for $P_m > P_e$ and $P_m < P_e$.
1:15–2:30	PowerFactory: initialize RMS simulation, select variables, create result/plot objects and run undisturbed simulation.
2:30–2:50	Inspect whether the model remains near its initial operating point.
2:50–3:00	Record any initialization/model errors and their causes.

11.2 Steady state versus dynamics

A load flow provides one operating point:

$$\frac{dx}{dt} = 0$$

for all dynamic state variables conceptually at equilibrium.

A dynamic model introduces state equations:

$$\dot{x} = f(x, u, t),$$

and algebraic network equations:

$$0 = g(x, y, u, t),$$

where x represents dynamic states, y algebraic variables and u inputs/disturbances.

PowerFactory RMS simulation numerically integrates the dynamic states while solving the network constraints.

11.3 Synchronous-machine energy balance

The turbine/prime mover supplies mechanical power P_m. The generator delivers electrical power P_e.

If

$$P_m = P_e$$

and losses are neglected, rotor speed remains steady.

If

$$P_m > P_e,$$

the excess mechanical power accelerates the rotor.

If

$$P_m < P_e,$$

the rotor decelerates.

11.4 Swing equation

A common simplified form is

$$M\frac{d^2\delta}{dt^2} = P_m - P_e,$$

or in per-unit forms involving inertia constant H and synchronous speed. The key physical meaning is more important than the exact convention:

- δ: rotor electrical angle relative to a synchronous reference;
- $P_m - P_e$: accelerating power;
- inertia: resistance to rapid speed change.

11.5 Frequency

System frequency is a collective indicator of active-power balance. If generation is suddenly lost, electrical demand temporarily exceeds mechanical input and system frequency tends to decline. Governors and other frequency-response resources act to rebalance power.

11.6 Stability categories

11.6.1 Rotor-angle stability

Ability of synchronous machines to remain in synchronism.

11.6.2 Frequency stability

Ability to maintain/restore frequency following severe active-power imbalance.

11.6.3 Voltage stability

Ability to maintain acceptable voltages as the system and controls respond to disturbance or stress.

11.6.4 Small-signal versus transient stability

Small-signal stability concerns behaviour following small disturbances and oscillatory modes. Transient stability concerns large disturbances such as faults, line trips and generator outages.

11.7 Dynamic controls

A realistic generator model may include:

- synchronous-machine electrical model;
- automatic voltage regulator (AVR);
- excitation system;
- turbine governor;
- power system stabiliser (PSS).

Do not attempt to tune these controls in Week 9. First understand the architecture.

11.8 PowerFactory practical lab – initialise and run an undisturbed RMS simulation

PowerFactory task

Task. Prepare a dynamic Study Case, use a valid synchronous-machine dynamic Type-/model for G1, calculate initial conditions, define recorded variables, run 10 s with no disturbance and verify the trajectories remain at equilibrium.

11.8.1 Task 1 – create a dedicated RMS Study Case

Open the Data Manager and go to `Study Cases`. Duplicate your working load-flow Study Case or create a new Study Case named `W09_RMS_NoDisturbance`. Activate it. Activate `OP_Normal` as the Operation Scenario.

Run a normal Load Flow first and verify that the operating point is sensible.

11.8.2 Task 2 – provide G1 with a dynamic machine model

Open `G1 -> Basic Data -> Type`. A machine Type created only for Week 4 steady-state work may not contain credible RMS parameters.

For this course, use a **standard synchronous-machine Type/model from the DIgSILENT/global library or a supplied PowerFactory example/template** with a rating reasonably close to the educational 50-MVA generator, then copy it into the project if you want to edit it. Adjust only clearly understood rated quantities to match the project.

Important modelling note

Do not invent transient/subtransient reactances and time constants simply to make the RMS command run. Dynamic stability is extremely sensitive to these parameters. The click-by-click workflow can be taught with a standard library model; a professional study requires manufacturer or validated planning-model data.

The DIgSILENT introductory RMS workflow explicitly includes use of standard models/templates, calculation of initial conditions, result-variable definition, events and graphical result visualisation.

11.8.3 Task 3 – calculate initial conditions

Switch to the **Simulation RMS/EMT** toolbox or open the command named **Calculation of Initial Conditions** (`ComInc`).

On its Basic Options select:

- **Method of Simulation**: RMS;
- **Network Representation**: balanced / positive sequence for this first course simulation;
- **Verify Initial Conditions**: enabled, if available.

Click **Execute**. Read the Output Window carefully. You want a successful initialization message and no unresolved model/initial-condition errors.

If it fails:

1. first rerun Load Flow and resolve any steady-state warning;
2. confirm the G1 dynamic model is in service and properly connected;
3. remove/disable optional controllers one at a time to isolate the failing model;
4. execute Initial Conditions again.

11.8.4 Task 4 – define result variables

Right-click `G1` on the SLD and choose `Define -> Results for Simulation RMS/EMT`. Select the active simulation result file/object. Add the available variables corresponding to:

- active power P;
- reactive power Q;
- rotor speed / speed deviation / electrical frequency;
- rotor angle (if exposed by the selected machine model).

Right-click `Bus33B -> Define -> Results for Simulation RMS/EMT` and add:

- voltage magnitude;

- frequency, if available at the terminal in your result set.

Variable names can differ by model. Use the variable description in the selection dialog rather than guessing from the internal code.

11.8.5 Task 5 – create plots

Use `Insert Plot` / the plot toolbar and insert one or more **Curve Plots**. Choose the active simulation result file as data source and add curves for:

1. Bus33B voltage;
2. G1 active power;
3. G1 speed/frequency;
4. G1 rotor angle, if available.

11.8.6 Task 6 – run with no disturbance

Open **Run Simulation** (`ComSim`). Set:

- start time = 0 s (or keep the initialized default);
- final time = **10 s**;
- RMS simulation method as already prepared by `ComInc`.

Click **Execute**, then refresh/open the plots.

Expected result: voltage, machine P/Q, speed and rotor angle should remain essentially constant, aside from negligible numerical settling. If a large oscillation appears with no event, treat it as a model/initialization problem to investigate before Week 10.

What to verify before moving on

Your deliverable is one 10-s no-disturbance plot and a statement that the initial load-flow operating point is dynamically consistent. If it is not, list the exact Output Window message and model that prevents successful initialization rather than continuing to fault simulations.

End-of-week checkpoint

Explain why a no-disturbance dynamic simulation is a valuable validation test and why dynamic simulation cannot simply start from arbitrary generator state values.

Chapter 12

Week 10 – Transient Stability, Fault Events and Clearing Time

Learning outcomes

Understand how a fault changes electrical power output, why the rotor accelerates, how clearing time affects stability, and how to build and interpret an RMS fault simulation.

12.1 Three-hour breakdown

0:00–0:50	Theory: fault-on power transfer, accelerating power, post-fault recovery and critical-clearing-time concept.
0:50–1:00	Sketch expected $V(t)$, speed and rotor-angle responses.
1:00–2:40	PowerFactory: create fault/clearing events, run RMS cases at multiple clearing times, plot results.
2:40–2:50	Determine qualitative stability boundary.
2:50–3:00	Write a physical explanation of one stable and one unstable case.

12.2 What happens during a severe fault

Before the fault,

$$P_m \approx P_e.$$

A nearby three-phase fault depresses voltage and can greatly reduce electrical power transfer:

$$P_e \downarrow .$$

Mechanical turbine power cannot change instantaneously:

$$P_m \approx \text{constant over the first instants.}$$

Therefore

$$P_a = P_m - P_e > 0$$

and the rotor accelerates. Rotor angle increases relative to the system reference.

12.3 After clearing

When protection clears the fault, the network changes to a post-fault condition. If sufficient synchronising power is restored, the machine can decelerate and remain in synchronism. If the fault lasts too long or the post-fault network is too weak, rotor angle may continue diverging.

12.4 Critical clearing time

The critical clearing time is the maximum approximate fault duration for which the considered system remains transiently stable under specified conditions. It is not a universal number for a bus; it depends on:

- pre-fault dispatch;
- fault location/type;
- clearing action;
- post-fault topology;
- machine inertia;
- controls;
- system strength.

12.5 Equal-area intuition

For a simplified single-machine infinite-bus system,

$$P_e = P_{\max} \sin \delta.$$

During the fault, $P_{\max}$ may fall sharply. The machine accumulates kinetic energy as it accelerates. After clearing, electrical power may exceed mechanical power over part of the trajectory, allowing deceleration. The equal-area criterion expresses this energy balance geometrically for simplified cases.

You do not need to perform detailed equal-area calculations in this course, but this picture explains why faster protection clearing improves transient stability.

12.6 PowerFactory practical lab – apply and clear a three-phase fault in RMS

PowerFactory task

Task. Duplicate the validated Week 9 dynamic case, create a fault-on event and a separate clearing event, run the RMS simulation and sweep the clearing time to observe increasing rotor-angle stress.

12.6.1 Task 1 – duplicate the validated RMS case

In `Data Manager -> Study Cases`, duplicate `W09_RMS_NoDisturbance` and rename the copy `W10_RMS_Fault`. Activate it. Keep `OP_Normal` active.

Execute Load Flow and then **Calculation of Initial Conditions**. Do not proceed until initialization is successful.

12.6.2 Task 2 – create the fault-on event

Use either of these interface paths:

- right-click the target terminal, e.g. `Bus110B`, and choose `Define -> Short-Circuit Event`; or
- in the Data Manager expand the active Study Case → **Simulation Events/Fault** and create a new **Short-Circuit Event** (`EvtShc`).

Set:

Event name	`Fault_ON`
Execution time	1.000 s
Target	Bus110B
Fault type	3-phase short circuit
Fault resistance	0 Ω
Fault reactance	0 Ω
Event action	apply / create short circuit

12.6.3 Task 3 – create the clearing event

A dynamic short circuit is not cleared merely because the simulation continues. Create a **second Short-Circuit Event** at the same target:

Event name	`Fault_CLEAR`
Execution time	1.100 s
Target	Bus110B (same location)
Event action / fault type	Clear Short-Circuit

Depending on the release, the clearing action may be selected through an event-action dropdown or a short-circuit type/action option. The essential configuration is a second `EvtShc` at the same location that clears the previously applied fault.

12.6.4 Task 4 – confirm result variables

Keep the Week 9 variables and make sure the active result object contains at least:

- Bus110B or Bus33B voltage;
- G1 active power;
- G1 rotor speed/frequency;
- G1 rotor angle.

12.6.5 Task 5 – initialize and run

After changing events, execute **Calculation of Initial Conditions** again. Then open **Run Simulation** and set final time to **5 s** or **10 s**. Execute.

Inspect the plots around $t = 1.0$ s:

1. voltage should drop sharply during the fault;
2. generator electrical P should reduce/change during the fault;
3. rotor speed should respond to the accelerating-power imbalance;
4. after clearance, voltage and electrical P recover and the rotor-angle/speed trajectory shows whether the machine resynchronises.

12.6.6 Task 6 – clearing-time sweep

Keep `Fault_ON` at 1.000 s. Change only `Fault_CLEAR` execution time and repeat initialization + simulation for:

`Fault_CLEAR` time	Fault duration	Qualitative response
1.080 s	80 ms	
1.120 s	120 ms	
1.200 s	200 ms	
1.300 s	300 ms	

For each run, record whether rotor-angle oscillations are bounded/settling or diverging, and record the largest observed angle/speed deviation if practical.

If a long-duration case becomes numerically or physically unstable, that is a result; do not force convergence by arbitrarily changing solver settings.

12.6.7 Task 7 – explain one stable trace

Annotate one plot with these intervals:

1. **0–1.0 s**: pre-fault equilibrium;
2. **fault-on**: voltage/electrical transfer depressed, machine accelerates if $P_m > P_e$;
3. **post-clearance**: synchronising/decelerating action and oscillatory recovery;
4. **settled or unstable**: state your observation.

What to verify before moving on

Show that changing only the clearing-event time changes the dynamic severity. Your explanation must connect the plot to accelerating power and rotor-angle behaviour, not merely describe visual oscillations.

End-of-week checkpoint

Explain an RMS fault plot without saying only "the signal oscillates". Link each stage to pre-fault balance, fault-on voltage/power reduction, rotor acceleration, clearing and post-fault synchronising/decelerating behaviour.

Chapter 13

Week 11 – Renewable Integration, Reverse Power Flow and Reactive-Power Control

Learning outcomes

Understand the steady-state impact of inverter-based generation on active/reactive flow, voltage, thermal loading, reverse power flow and grid support.

13.1 Three-hour breakdown

0:00–0:50	Theory: inverter-based generation, P/Q capability, voltage rise, reverse flow and common reactive-control modes.
0:50–1:05	Hand reasoning: local generation versus local demand and direction of upstream MW.
1:05–2:35	PowerFactory: add 30-MW PV representation; sweep output; compare PF, fixed Q and voltage control where available.
2:35–2:50	Experiment: minimum load + maximum PV.
2:50–3:00	Summarise limiting issues and possible controls.

13.2 How inverter-based generation differs

A synchronous generator directly couples a rotating electrical machine to the AC system. A PV plant interfaces through power electronics:

$$\text{PV array} \to \text{DC link} \to \text{inverter} \to \text{transformer} \to \text{AC grid}.$$

For steady-state load flow, many internal fast controls can be abstracted. The principal quantities are:

$$P, \quad Q, \quad V, \quad \text{current/MVA capability}.$$

For dynamic studies, however, inverter controls and plant controller models become essential.

13.3 Reverse power flow

Suppose a local feeder has:

$$P_{\text{load}} = 20 \text{ MW}.$$

If PV produces

$$P_{\text{PV}} = 10 \text{ MW},$$

approximately 10 MW still arrives from upstream, neglecting losses.

If

$$P_{\text{PV}} = 30 \text{ MW},$$

local generation exceeds local demand and approximately 10 MW exports upstream. This is reverse power flow relative to the traditional one-way distribution assumption.

Consequences can include:

- changed line/transformer loading;
- voltage rise;
- changed protection current directions;
- altered tap-controller behaviour;
- changed losses;
- export constraints.

13.4 Voltage-rise intuition

The approximate relation

$$\Delta V \approx \frac{RP + XQ}{V}$$

must be interpreted with the adopted sign convention. Reversing active-power flow can reverse the direction of the active-power contribution to voltage drop. Therefore generation near the end of a feeder can raise local voltage.

13.5 Reactive-control modes

Common concepts are:

13.5.1 Unity/fixed power factor

The inverter follows a specified power factor. At PF=1,

$$Q \approx 0.$$

13.5.2 Fixed Q

The plant injects or absorbs a specified MVAr.

13.5.3 Voltage control

The plant changes Q to regulate a controlled bus voltage, subject to reactive/current limits.

13.5.4 Q(V) or Volt-VAr characteristic

Reactive power changes as a function of voltage according to a droop/characteristic. This avoids one controller trying to hold an infinitely stiff voltage target.

13.6 Capability limits

An inverter with apparent-power rating $S_{\max}$ approximately satisfies

$$P^2 + Q^2 \leq S_{\max}^2$$

subject to more detailed converter, voltage and grid-code limits. At high active power, available reactive capability may be reduced if the converter is MVA/current limited.

13.7 PowerFactory practical lab – connect a 30-MW PV plant and study P/Q control

PowerFactory task

Task. Add a Static Generator as the steady-state PV-plant representation, sweep its MW output, compare reactive-control modes and create a low-load/high-PV reverse-power-flow case.

13.7.1 Task 1 – place the PV plant

From the Drawing Toolbox choose **Static Generator** (`ElmGenstat`) and connect it to `Bus33B`. Open its dialog and set:

Name	`PV_30MW`
Connection	Bus33B, 33 kV
Number of parallel units	1, unless your model uses a plant aggregation field
Educational apparent-power capability	33 MVA
Initial active power	0 MW
Initial reactive power	0 MVAr

Depending on the model, apparent-power/rating data may be on Basic Data, a Type or the short-circuit/RMS model. Enter it only in the location expected by your selected Static Generator variant.

Create and save an Operation Scenario named `OP_PV_Normal`.

13.7.2 Task 2 – set constant-Q / unity-PF operating mode

Open `PV_30MW -> Load Flow`. Choose a local reactive control mode that keeps Q fixed (often **Const. Q**) or choose the **P,Q** input mode. Set:

$$Q = 0 \text{ MVAr}.$$

If the model instead uses **P, cos(phi)**, set PF = 1.00. Set educational reactive limits to about $-10, +10$ MVAr where the model permits limits.

13.7.3 Task 3 – active-power sweep

With LoadA/B/C at the normal values, set PV active power to:

$$0, ightarrow10, ightarrow20, ightarrow30 \text{ MW}.$$

At every point execute Load Flow and record:

PV MW	Bus33B V	T2 P	T2 loading	GridSource P	Losses
0					
10					
20					
30					

Pay attention to the sign/direction arrow for T2 active power. As local PV output increases, upstream import should reduce; if local generation exceeds downstream demand, the direction can reverse.

13.7.4 Task 4 – compare three reactive-control conditions at 30 MW

Keep PV P = 30 MW.

Case A – unity PF / Q = 0. Set constant Q = 0 MVAr. Execute and record Bus33B voltage.

Case B – reactive absorption. Set constant Q so the PV plant absorbs approximately 5 MVAr. In many PowerFactory generator conventions this is $Q = -5$ MVAr, but verify the result/arrow in your version. Execute and record Bus33B voltage.

Case C – reactive injection. Set Q so the plant injects approximately +5 MVAr; execute and record voltage.

Case D – voltage control, if available. Change the local reactive controller to **Const. V** / Voltage Control, set voltage target = **1.00 pu**, and retain Q limits near $-10, +10$ MVAr. Execute and record the solved Q required to regulate the bus.

Compare the four cases. Do not rely only on the sign of the Q input; confirm physically whether the plant is supplying or absorbing reactive power from the result arrows/values.

13.7.5 Task 5 – low-load/high-PV reverse-flow scenario

With the 30-MW PV in service, create `OP_LowLoad_HighPV` using `File -> Save Operation Scenario as...`. Set:

LoadA	10 MW	PF 0.95
LoadB	5 MW	PF 0.92
LoadC	10 MW	PF 0.97
PV	30 MW	Q = 0 initially

Execute Load Flow. Inspect active-power arrows/values through L33, T2 and the 110-kV network. Record Bus33B voltage and identify the first location where the active-power direction is opposite to the no-PV case.

If Bus33B rises above the educational 1.05-pu criterion, repeat with 5 MVAr reactive absorption or voltage control and quantify the improvement.

What to verify before moving on

Your final evidence should show one case of reduced upstream import and, if the chosen load levels allow it, one case of reverse active-power flow. Explain why the low-load/high-PV case can be more critical for overvoltage than the peak-load case.

End-of-week checkpoint

Explain why the "maximum demand" case is not automatically the worst case after distributed generation is connected. High generation and low demand can produce overvoltage and reverse-flow constraints.

Chapter 14

Week 12 – Final Integrated Grid-Connection Study

Learning outcomes

Independently perform a small planning study for a proposed 30-MW solar connection and produce an engineering recommendation supported by load flow, voltage-control, short-circuit and $N-1$ evidence.

14.1 Three-hour breakdown

0:00–0:30	Define study assumptions, scenarios, acceptance criteria and monitored quantities.
0:30–2:25	Perform base, PV, operating-extreme, short-circuit and contingency calculations.
2:25–2:45	Compare cases and identify the binding constraint.
2:45–3:00	Write a concise engineering recommendation and list required follow-up studies.

14.2 Study question

A new 30-MW solar plant is proposed at Bus33B.

Your task is to answer:

> Can the proposed plant connect without unacceptable voltage, thermal, fault-level or $N-1$ impacts under the studied conditions, and if not, what mitigation should be investigated?

14.3 PowerFactory practical lab – execute the complete connection study

PowerFactory task

Task. Use the project you have built over eleven weeks to execute a small, traceable grid-connection study for `PV_30MW`. You will create named scenarios, run the calculations in a fixed order, fill a results matrix and write a recommendation.

14.3.1 Task 1 – create the study workspace and scenarios

Duplicate a clean Study Case and rename it `W12_ConnectionStudy`. Activate it. Make sure the network is restored to the baseline topology: L1, L2, L33, T1, T2 and G1 in service; temporary T2B removed/out of service; capacitor state explicitly known.

Create/save the following Operation Scenarios. Each time, use `File -> Save Operation Scenario as...`, enter the stated values, and then `File -> Save Operation Scenario`.

Scenario	Load scale	PV P	PV Q mode	Purpose
`BASE_NoPV`	normal	0 MW	Q=0	pre-connection reference
`PV_Normal`	normal	30 MW	Q=0	normal generation
`PV_HighLoad`	1.4 x normal	30 MW	Q=0	thermal / low-V test
`PV_LowLoad`	0.5 x normal	30 MW	Q=0	export / high-V test

For 0.5 x normal use approximately LoadA = 12.5 MW, LoadB = 7.5 MW and LoadC = 10 MW with their original PF values. For 1.4 x normal use 35, 21 and 28 MW respectively.

14.3.2 Task 2 – define the educational acceptance criteria

Write these at the top of your results sheet:

$$0.95 \leq V \leq 1.05 \text{ pu},$$

$$\text{continuous line/transformer loading} \leq 100\%.$$

For short circuit, do not invent a pass/fail breaker rating. Record the calculated change in fault level and state that equipment acceptance requires the actual switchgear making/breaking and thermal ratings.

14.3.3 Task 3 – run the base case

Activate `BASE_NoPV`.

1. Open **Calculate Load Flow** and execute.
2. Record Bus33A/Bus33B voltage, maximum line loading, T2 loading, GridSource MW/MVAr and losses.
3. Right-click `Bus33B -> Calculate -> Short-Circuit`.
4. Method = IEC 60909; Fault Type = 3-phase; maximum case; Execute.
5. Record I_k'', S_k'' and peak current if displayed.

This row becomes the reference against which all PV results are compared.

14.3.4 Task 4 – run the normal 30-MW PV case

Activate `PV_Normal`. Execute Load Flow using the same `ComLdf` options. Record the same metrics and calculate:

$$\Delta V_{33B} = V_{33B,PV} - V_{33B,base},$$

$$\Delta P_{Grid} = P_{Grid,PV} - P_{Grid,base}.$$

Inspect T2's active-power direction so you can state whether import has merely reduced or actually reversed.

14.3.5 Task 5 – run high-load and low-load extremes

Activate `PV_HighLoad`, execute Load Flow and record the same metrics. Identify whether the limiting concern is low voltage or thermal loading.

Then activate `PV_LowLoad`, execute and record. Identify whether Bus33B approaches/exceeds 1.05 pu and whether T2 exports toward 110 kV.

If `PV_LowLoad` creates high voltage, test one mitigation without overwriting the scenario permanently:

1. open `PV_30MW -> Load Flow`;
2. change from Q=0 to approximately 5 MVAr reactive absorption (verify sign from results), *or* use voltage control at 1.00 pu with Q limits;
3. execute Load Flow;
4. record the new Bus33B voltage and PV Q;
5. restore the saved `PV_LowLoad` scenario afterward.

14.3.6 Task 6 – compare short-circuit level before and after PV

Run the same Bus33B IEC 60909 three-phase fault in `BASE_NoPV` and `PV_Normal`. Record both values in adjacent cells.

Important modelling note

A Static Generator's fault contribution depends on the short-circuit model and parameters configured for that generator. If you have not entered validated inverter short-circuit current parameters, do not interpret the numerical difference as a manufacturer-accurate PV fault contribution. State explicitly which model/assumption PowerFactory used. Inverter fault current must not be assumed equal to that of a synchronous generator of the same MW rating.

14.3.7 Task 7 – perform the critical $N-1$ checks

Choose the more stressful PV operating scenario based on your results—usually `PV_HighLoad` for thermal/low-voltage constraints or `PV_LowLoad` for export/high-voltage constraints.

Perform manually:

1. **L1 outage**: L1 Out of Service → execute Load Flow → record L2 loading and min/max voltage → restore L1.
2. **L2 outage**: L2 Out of Service → execute → record L1 loading and voltages → restore L2.
3. **T2 outage**: T2 Out of Service → execute → record whether the 33-kV network becomes unsupplied/islanded → restore T2.

Always restore the previous contingency before applying the next. If automated Contingency Analysis is licensed, run it afterward as a cross-check.

14.3.8 Task 8 – fill the final study matrix

Case	Bus33B V	Max line %	T2 %	Grid P	Losses	Bus33B I_k''	Pass/issue
BASE No PV							
PV Normal							
PV High Load							
PV Low Load							
PV Low Load + Q mitigation						–	
Critical N-1						–	

14.3.9 Task 9 – write the recommendation

Write four short paragraphs:

1. **Finding**: state whether normal cases meet the educational voltage/loading criteria.
2. **Worst condition**: identify the scenario/contingency causing the largest issue.
3. **Electrical cause**: link the issue to reactive demand, reverse flow, impedance, loss of a parallel path or another observed physical cause.
4. **Action**: state the mitigation/follow-up study required.

A good conclusion is specific, for example:

> The 30-MW connection is acceptable in the studied normal-demand case, but Bus33B exceeds the educational upper-voltage criterion during the low-demand/high-generation condition. The behaviour is driven by export through the 33-kV feeder and associated voltage rise. Reactive-power absorption or voltage control reduces the voltage in the training model; therefore plant reactive capability, transformer control and any export/curtailment requirement should be assessed using utility-approved network and inverter models before final connection approval.

14.4 What not to overclaim

A 12-week educational model is not a bankable grid-connection study. A professional study can require:

- validated utility network models;
- official planning criteria and outage lists;
- detailed inverter vendor models;
- RMS and potentially EMT studies;
- harmonic and flicker assessment;
- protection review;
- reactive-power capability assessment;
- grid-code compliance studies;
- multiple seasonal and dispatch conditions.

End-of-week checkpoint

You have completed the programme successfully if you can start from a blank study question, select the required PowerFactory calculations, build/validate operating cases, interpret results physically and state a defensible conclusion with appropriate limitations.

Chapter 15

Appendix A – Formula Sheet

15.1 Three-phase power

$$S = P + jQ$$

$$|S| = \sqrt{P^2 + Q^2}$$

$$P = \sqrt{3} V I \cos\phi$$

$$Q = \sqrt{3} V I \sin\phi$$

$$|S| = \sqrt{3} V I$$

$$\mathrm{PF} = \frac{P}{|S|}$$

$$Q = P \tan(\cos^{-1} \mathrm{PF})$$

15.2 Loss and voltage-drop intuition

$$P_{\text{loss}} = 3I^2 R$$

$$\Delta V \approx \frac{RP + XQ}{V}$$

15.3 Per-unit

$$I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3} V_{\text{base}}}$$

$$Z_{\text{base}} = \frac{V_{\text{base}}^2}{S_{\text{base}}}$$

$$Z_{\mathrm{pu}} = \frac{Z}{Z_{\mathrm{base}}}$$

$$Z_{\%} = 100 Z_{\mathrm{pu}}$$

15.4 Short circuit

$$I_f \approx \frac{V_{\mathrm{th}}}{Z_{\mathrm{th}} + Z_f}$$

$$S_k'' = \sqrt{3} V_n I_k''$$

$$i_p \approx \kappa \sqrt{2} I_k''$$

15.5 Dynamics

$$P_a = P_m - P_e$$

$$M \frac{d^2\delta}{dt^2} = P_m - P_e$$

For a simplified single-machine infinite-bus transfer:

$$P_e = P_{\max} \sin\delta.$$

Chapter 16

Appendix B – Glossary

AVR
: Automatic Voltage Regulator; controls generator excitation to regulate voltage/reactive behaviour.

Bus / Terminal
: Electrical node at which connected equipment shares a voltage.

Contingency
: Credible outage or disturbance investigated in a security study.

External Grid
: Equivalent representation of a larger network/source, often used as a voltage-angle reference and fault-strength equivalent.

Fault Level
: Magnitude of short-circuit strength/current at a location under specified conditions.

Inertia Constant H
: Stored kinetic energy of a rotating machine at rated speed divided by its MVA rating, commonly expressed in seconds.

Load Flow
: Steady-state AC network solution for voltages, angles and power flows.

OLTC
: On-load tap changer used to alter transformer ratio while energised.

PQ Bus
: Bus where active and reactive power are specified and voltage magnitude/angle are solved.

PV Bus
: Bus where active power and voltage magnitude are specified and reactive power/angle are solved, subject to limits.

Per Unit
: Normalised representation relative to selected electrical bases.

Reactive Power Q
: Imaginary component of complex power associated with alternating electric/magnetic field energy.

RMS Simulation
: Time-domain simulation using RMS/phasor representations suited to electromechanical and control dynamics.

Slack Bus
: Reference bus where voltage magnitude and angle are specified and active/reactive power balance is solved.

Study Case
: PowerFactory container/context used to manage a reproducible calculation state and associated commands/results.

Type
: Reusable object holding shared equipment characteristics.

Zero Sequence
: Symmetrical-component set with all three phase components in phase; strongly linked to earth/neutral return paths.

Chapter 17

Appendix C – Recommended Study Discipline

17.1 The prediction habit

Before every PowerFactory calculation, write one or two expected trends. Examples:

- If load MW rises, current should rise.
- If power factor worsens at fixed MW, MVAr and current should rise.
- If source impedance increases, short-circuit current should fall.
- If a parallel line trips, remaining paths should carry more power.
- If local PV exceeds local load, upstream active-power flow may reverse.

Then calculate. If the result contradicts your prediction, do not immediately assume the software is wrong. Investigate the model, control modes, sign conventions and topology.

17.2 The four-column notebook

Maintain one table throughout the course:

Concept	Equation	PowerFactory setting	Engineering interpretation
Reactive demand	$Q = P\tan(\cos^{-1}\mathrm{PF})$	Load PF / Q input	More Q generally increases current and voltage drop
Fault strength	$I_f \sim V/Z_{\mathrm{th}}$	External-grid short-circuit data	Stronger source produces higher fault duty

17.3 A model-validation checklist

Before trusting a result, check:

1. Correct nominal voltages?
2. Correct equipment connectivity?
3. Correct in/out-of-service states?

4. Plausible line/transformer impedance?
5. Plausible equipment ratings?
6. Correct load MW/MVAr or PF?
7. Correct generator active-power and voltage/reactive mode?
8. Appropriate external-grid strength?
9. Correct transformer tap/grounding/vector group for the study?
10. Appropriate calculation method and limits?
11. Do directions and magnitudes make physical sense?

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